

AI Hiring Assistant

1. What Are We Building?

A website that helps companies hire faster and more fairly using AI. Instead of a recruiter reading hundreds of resumes manually, the AI reads them automatically, scores each candidate, and explains every decision. **Recruiters always stay in control** — they can accept, reject, or override any AI decision at any time.

Problem Today	Our Solution
Hours spent reading resumes manually	AI reads every resume in minutes
Good candidates missed; inconsistent decisions	Every resume scored with clear explanation
Candidates wait weeks without updates	Automatic email updates at every stage

2. Business Goals

Goal	What Success Looks Like
Save recruiter time	Reduce manual resume reading by 70%
Shortlist faster	Shortlist ready in under 1 hour
AI–recruiter alignment	Recruiter agrees with AI ranking 85% of the time
Faster hires	Reduce total hiring time by 30%
Auto interview booking	90% of interviews booked automatically
Candidate satisfaction	Score of 4.2 / 5.0
Full audit trail	Every action saved and reviewable
Scale	Support 100+ companies on the platform

3. Who Uses This?

Role	Title	Key Needs
Recruiter / HR Staff	Recruiter / HR Staff	See AI rankings, understand scores, book interviews, send offers
Job Applicant	Job Applicant	Apply easily, track status in real time, receive feedback
HR Manager	HR Manager / Approver	Approve offers, view hiring reports, review decisions
Admin / IT	Admin / IT	Manage user access, monitor system, export activity logs

4. How AI Screening Works

Step	Process Stage	Description
1	Job Requisition Published	The recruiter submits structured job requirements. The system records and stores all role parameters for screening.
2	Application Submission	The candidate submits a resume in PDF or Word format. The system performs automated extraction of skills, experience, and academic qualifications.
3	AI Profile Analysis	The AI engine evaluates the extracted candidate profile against the defined job requirements using weighted matching criteria.
4	Optimal Candidate Identification	The system identifies and prioritises candidates whose profiles demonstrate the highest degree of alignment with the role specifications.
5	Scoring & Assessment Report	Each candidate is assigned a composite score (0–100) along with a structured assessment of competency strengths, gaps, and a shortlisting recommendation.
6	Candidate Classification	All applicants are ranked by score and categorised into three tiers: Group A (recommended), Group B (under consideration), Group C (not progressed).
7	Recruiter Review & Decision	The recruiter reviews the ranked candidates alongside AI-generated assessments and retains full authority to accept, modify, or override any classification.
8	Automated Communications & Audit Logging	System-generated notifications are dispatched to candidates at each decision point. All actions are recorded in the immutable activity log.

Scoring: Points given for matching skills, years of experience, education level, and overall fit. Each factor is weighted (skills weighted highest). Recruiters can adjust weights per job. **Safety:** AI never auto-rejects silently — uncertain profiles are flagged for manual review.

5. Features We Are Building

Feature Area	What the App Does
Login & Accounts	Email/password or Google sign-in. Role-based access: applicant, recruiter, manager, admin.
Job Postings	Create, edit, copy, pause, or close listings. Fields: title, department, location, skills, experience, salary.
Resume Upload	PDF or Word, max 10 MB. Auto-extracts skills, work history, education into candidate profile.
Applications	One application per person per job. Status tracking: Applied → Shortlisted → Interview → Offer → Hired/Rejected.
AI Scoring	Score (0–100), skill & experience match %, strengths/weaknesses, written summary, recommendation.
Recruiter Controls	Group A/B/C tabs. Accept or override AI decisions with a reason. Bulk actions supported.
Automatic Emails	Application received, rejection (with feedback), shortlist, interview invite, reminder, offer, onboarding.
Interview Booking	Recruiter sets availability. App books via Google Calendar and creates Google Meet links automatically.
Job Offers	HR Manager creates offer using template → approval flow → digital acceptance by candidate.
Onboarding	Welcome email + task checklist (submit ID, fill forms, etc.) triggered on offer acceptance.
Admin Tools	Add/remove users, assign roles, view AI usage, search/export activity logs, update settings.

6. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + HTML + CSS	Browser screens and pages; fast interactive UI.
Backend	Node.js (JavaScript)	Server logic, business rules, API routing, database connections.
Database	MySQL	Stores user accounts, jobs, applications, scores, emails, interviews.
AI / ML	Gemini API (Google)	Reads resumes, compares to job requirements, generates scores and summaries.
File Storage	Google Cloud / AWS S3	Secure cloud storage for uploaded resumes and documents.
Email	SendGrid / Gmail API	Reliable delivery of all automated candidate and recruiter emails.
Calendar	Google Calendar + Meet	Auto-creates calendar events and video-call links for interviews.
Auth	Email/Password + Google Sign-In	Standard login or one-click Google OAuth for all user types.

7. Database Tables (MySQL)

Table	Data Stored
Users	Name, email, encrypted password, role, account status
Jobs	Title, department, location, required skills, experience, salary, status (draft/open/closed)
Candidate Profiles	Extracted resume data: skills, work history, education, certifications, projects
Applications	Candidate–job link, answers to screening questions, status, AI group (A/B/C)
AI Results	Score, skill/experience match %, strengths, weaknesses, summary, recommendation, AI version
Recruiter Decisions	Decision made, whether it overrides AI, reason given, feedback notes
Interviews	Date/time, Google Meet link, interviewer, status, post-interview feedback
Offers	Offer content, salary, approver, expiry date, candidate response
Onboarding	Checklist items, documents submitted, progress percentage
Activity Log	Who did what, when, and what changed — immutable; cannot be deleted or edited

8. Permissions by Role

Action	Applicant	Recruiter	HR Manager	Admin
Create account / manage own profile	✓	—	—	—
Apply for jobs / accept or decline offers	✓	—	—	—
See own application status	✓	—	—	—
Create and manage job postings	—	✓	✓	✓
See AI scores and candidate rankings	—	✓	✓	✓
Accept, reject, or override AI decisions	—	✓	✓	—
Book interviews / send offer letters	—	✓	✓	—
Approve offer letters	—	—	✓	✓
View hiring reports	—	Own jobs	Full team	Everyone
Manage user accounts and roles	—	—	—	✓
View full activity log	—	—	Limited	✓

9. Performance & Reliability

Expectation	Target
Page load speed	Any page loads in under 2 seconds
AI scoring speed	1 candidate in <15 s; 100 candidates in <5 min
Concurrent users	Up to 1,000 simultaneous users
Data volume	100,000 candidate profiles; 10,000 jobs
App uptime	99.5% availability (<4 hours downtime/month)
Failed actions	Auto-retry; failed items saved for review
AI unavailable	Application flagged for manual review — never silently skipped

10. Security & Privacy

Area	What We Do
Passwords	Encrypted with a secure one-way hash — never stored as plain text.
Data in transit	All browser-to-server traffic encrypted (TLS/HTTPS).
Data at rest	Database and file storage fully encrypted.
Privacy (GDPR)	Users can request data deletion. Minimal collection. Each company's data isolated.
Access control	Users see only what their role allows; no cross-company data access.
Suspicious activity	Too many failed logins locks the account; unusual activity is flagged.
AI fairness	AI never sees age, gender, or nationality. All decisions explained and logged.
File uploads	Resumes scanned for safety before storing in secure cloud.

11. Risks & Mitigations

Risk	How We Reduce It
Biased AI scores	AI never sees protected attributes. All decisions logged. Recruiter can always override.
AI hallucination	AI must cite specific resume evidence for every score point. Outputs validated for format.
Resume misread	Candidate can review and edit extracted data before submitting.
Over-reliance on AI	Group C cannot be auto-rejected — recruiter must confirm. Every override requires a reason.
Data breach	Full encryption, strict access controls, regular security testing, GDPR compliance.
AI service outage	App falls back to manual mode. Auto-retry. Critical jobs have a backup flow.
Scope creep	Fixed 5-week MVP scope. New features go to backlog, reviewed post-launch.
Low recruiter adoption	Plain-English AI explanations. Recruiters involved in UAT. Training provided.

12. Build Plan & Phases

Phase 1 — MVP (Sub-phases 1.1–1.5)			Phase 2–4 — Post-Launch Enhancements	Phase 5 — Enterprise Expansion
Week	Phase	Sub-Phase	Weekly Deliverables	
Week 1	Phase 1 MVP	1.1 Frontend Development	Deliverables: Applicant portal, Recruiter portal, Admin portal screens built in React.js. Role-based navigation and authentication UI (email/password + Google sign-in). Responsive layout across all device sizes. Manager Checkpoint: All three portals visible and navigable in browser.	
Week 2	Phase 1 MVP	1.2 Backend & Database	Deliverables: Node.js REST API fully operational. MySQL database schema created and seeded. Resume upload endpoint live (PDF/Word, max 10 MB). Automated data extraction pipeline returning structured candidate profiles. Cloud storage integration for secure file handling. Manager Checkpoint: Resume uploaded → data extracted and stored in database.	
Week 3	Phase 1 MVP	1.3 AI Integration & Scoring	Deliverables: Gemini API connected and operational. Each submitted resume automatically scored (0–100). Skill match %, experience match %, strengths and gaps analysis, and written assessment summary generated per candidate. Candidates auto-classified into Group A / B / C. Manager Checkpoint: Upload resume → AI score and report visible in recruiter dashboard.	
Week 4	Phase 1 MVP	1.4 Recruiter Module, Emails & Interviews	Deliverables: Recruiter ranking dashboard (Group A/B/C tabs) live. Accept / reject / override decision workflow with mandatory reason field. Automated email notifications at each stage (applied, shortlisted, rejected, interview invite). Google Calendar and Google Meet integration — interview slots booked automatically. Offer letter generation and HR Manager approval workflow. Manager Checkpoint: Full recruiter flow end-to-end: score → decision → email → interview booked.	
Week 5	Phase 1 MVP	1.5 Admin, QA & Launch	Deliverables: Admin panel live — user management, role assignment, activity log viewer, system settings. Full end-to-end functional testing completed. Security audit and vulnerability check passed. Performance benchmarks validated (page load <2 s, AI scoring <15 s). Production deployment completed. Manager Checkpoint: System live in production — all roles tested and signed off.	
Wks 6–9	Phase 2	Enhanced AI & Analytics	Refined AI assessment narratives. Granular skill-gap detail in rejection communications. Recruiter analytics dashboard with applicant volume, shortlist rate, and time-to-hire metrics. AI model refinement based on accumulated recruiter override data.	
Wks 10–13	Phase 3	Interview Management	Multi-round and panel interview scheduling support. Candidate-initiated rescheduling and cancellation workflows. Structured post-interview feedback collection forms for interviewers.	
Wks 14–17	Phase 4	Offers & Onboarding	Multi-level offer approval workflow. Counter-offer and salary negotiation flow. New hire document collection and identity verification. Full onboarding task management and progress tracking.	
Future	Phase 5	Enterprise Expansion	Multi-tenancy architecture for enterprise clients. Diversity and inclusion reporting. Third-party HR system integrations (Workday, Greenhouse). Mobile application for candidates and recruiters. AI-assisted job description authoring.	